

Secret Sharing Scheme

Theoretical Overview

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Why Secret Sharing?

- Split a secret into *shares* so that only authorized subsets can reconstruct it [Bei11].
- Reduce single points of failure (key escrow, key backup, cloud storage, root keys).
- Foundation for threshold cryptography and secure multi-party computation.
- Used in practice for recovery and security hardening.

Motivation II

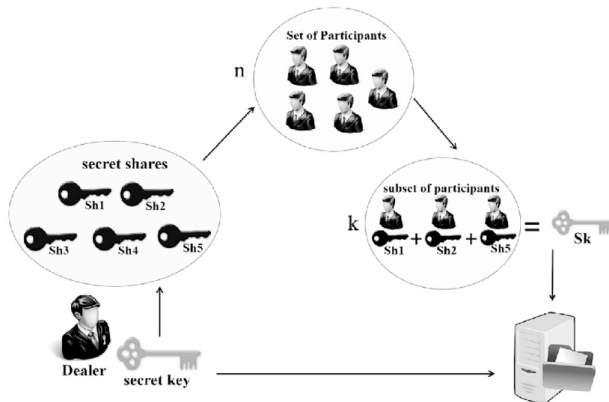


Figure: Threshold Secret Sharing intuition [GAQ20]

Foundations: Access Structures

- Parties: $\mathcal{P} = \{P_1, \dots, P_n\}$.
- Access structure $\mathcal{A} \subseteq 2^{\mathcal{P}}$: subsets that can reconstruct.
- **Monotone** requirement: if $A \in \mathcal{A}$ and $A \subseteq B$, then $B \in \mathcal{A}$ [Bei11].

Examples

- Threshold: "any t out of n ".
- Boolean policy: $(\text{CEO} \wedge \text{CFO}) \vee (2 \text{ of } 3 \text{ managers})$.
- Compartmented / hierarchical policies (common in organizations).

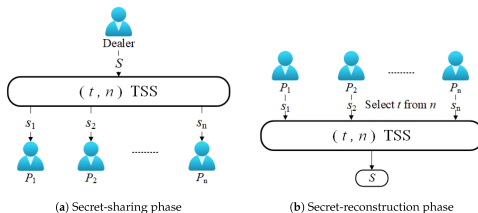


Figure: Secret Sharing Scheme Phases

Security Definitions and Metrics [Bei11]

Let secret S be random and shares be random variables $\{S_i\}$. A scheme realizing $\mathcal{A}$ satisfies:

- **Correctness:** for every $A \in \mathcal{A}$, S is determined by $\{S_i : P_i \in A\}$.
- **Perfect privacy:** for every $T \notin \mathcal{A}$, shares reveal no information:

$$I(S; \{S_i : P_i \in T\}) = 0 \iff H(S | \{S_i : i \in T\}) = H(S).$$

Efficiency metrics

- **Information rate:** share size vs secret size (ideal if equal).
- **Computation:** sharing + reconstruction complexity.
- **Robustness/verifiability:** tolerate or detect cheating.

Context: ideality

- **Ideal scheme:** share size equals secret size.
- Threshold structures are ideal; some non-threshold structures also are [Bri89].

Method 1: Shamir Threshold SSS [Sha79] I

Access structure: t -out-of- n

$$\mathcal{A}_t = \{A \subseteq \mathcal{P} : |A| \geq t\}.$$

Sharing idea

- Work over a finite field $\mathbb{F}_q$ (prime power), choose distinct nonzero points $x_1, \dots, x_n \in \mathbb{F}_q$.
- Sample random coefficients $\alpha_1, \dots, \alpha_{t-1} \leftarrow \mathbb{F}_q$.
- Define degree- $(t-1)$ polynomial:

$$P(x) = s + \alpha_1 x + \dots + \alpha_{t-1} x^{t-1}.$$

- Give share $(x_i, P(x_i))$ to party P_i .

Why it works

- Any t points interpolate a degree- $(t-1)$ polynomial $\Rightarrow$ reconstruct $s = P(0)$.

Method 1: Shamir Threshold SSS [Sha79] II

- Any $t - 1$ shares reveal no information about s (perfect privacy).
- Threshold schemes are often used as the baseline due to simplicity and efficiency.

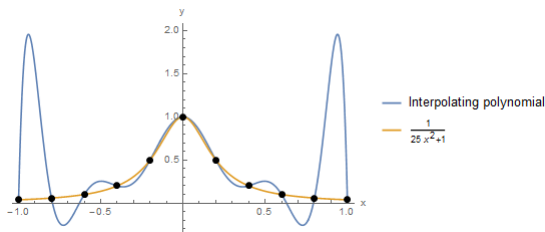


Figure: Lagrange Interpolation of $f(x) = \frac{1}{25x^2+1}$

Reconstruction (Lagrange Interpolation)

Given any authorized set $A = \{i_1, \dots, i_t\}$ with shares (x_{i_j}, y_{i_j}) where $y_{i_j} = P(x_{i_j})$,

$$s = P(0) = \sum_{j=1}^t y_{i_j} \cdot \lambda_{i_j}(0), \quad \lambda_{i_j}(0) = \prod_{\ell \neq j} \frac{0 - x_{i_\ell}}{x_{i_j} - x_{i_\ell}} \quad \text{in } \mathbb{F}_q.$$

Input: t pairs $(x_1, y_1), \dots, (x_t, y_t)$ in $\mathbb{F}_q^2$

Output: Secret $s = P(0)$

Compute $c_j \leftarrow \prod_{\ell \neq j} \frac{-x_\ell}{x_j - x_\ell}$ for each $j \in \{1, \dots, t\}$

Return $s \leftarrow \sum_{j=1}^t y_j \cdot c_j$

Algorithm 1: Shamir-Reconstruct(t shares)

Note: implementations often precompute coefficients for fixed x_i .

Why general access structures?

- Real policies are rarely "any t out of n ".
- Example: $(\text{CEO} \wedge \text{CFO}) \vee (3 \text{ of } 5 \text{ board members})$.
- Secret sharing can realize any monotone $\mathcal{A}$, but **efficiency** is the main challenge.

Key idea

- Describe $\mathcal{A}$ via a monotone formula / span program.
- Implement AND/OR composition by combining smaller schemes along the formula structure.

Current Methods 3: LSSS and CRT-based Schemes

- Many efficient constructions are **linear**: shares are linear combinations of secret + randomness over a field.
- A common abstraction uses:
 - a matrix M over $\mathbb{F}_q$,
 - a labeling ρ mapping each row to a participant.
- Authorized sets can reconstruct because their rows "span" the secret component; unauthorized sets cannot (perfect privacy) [Bei11].
- Another family: CRT-based secret sharing (shares are residues modulo carefully chosen integers) [Ift07].

Problem

- In real systems, participants might submit **incorrect** shares (accidental corruption or malicious cheating).
- Naive reconstruction can output a wrong secret without detection.

Typical goals

- **Verifiable Secret Sharing (VSS)**: parties can verify their shares are consistent with one committed secret.
- **Cheating-secure reconstruction**: detect cheaters or tolerate a bounded number of bad shares.

Trade-offs

- Adding robustness usually increases:
 - communication (extra proofs/commitments),
 - computation (verification),
 - share size (redundancy).
- Key design dimensions:
 - privacy threshold vs robustness threshold,
 - one-shot vs repeated reconstructions,
 - adversary model (passive vs active).

Variant	Privacy	Detect cheating	Overhead
Basic SSS	✓	–	low
VSS	✓	✓	medium
Cheating-secure	✓	✓	medium/high

Table: High-level comparison (conceptual).

Extension: Proactive Secret Sharing [HJKY95]

- Motivation: long-lived secrets face **perpetual leakage** (devices get compromised over time).
- Proactive SSS periodically **refreshes shares** without changing the underlying secret s .
- Security intuition: as long as the adversary does not control too many parties *within one refresh period*, the secret remains protected.

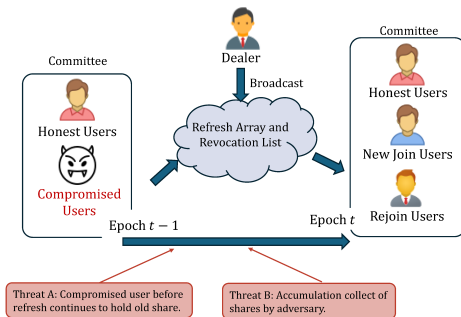


Figure: Proactive Secret Sharing - Refresh over Time

Extension: Quantum Secret Sharing (High-level) [Got00]

- Quantum secret sharing distributes a quantum state so only authorized sets can reconstruct it.
- Connects to quantum error-correcting codes: authorized sets can recover; unauthorized sets learn nothing.
- Relevance: shows the access-structure viewpoint generalizes beyond classical secrets.

- **Finite fields** $\mathbb{F}_q$: arithmetic for Shamir/LSSS (addition, multiplication, inverses).
- **Polynomial interpolation**: efficient reconstruction (Lagrange coefficients).
- **Linear algebra**: matrix-based secret sharing and reasoning about reconstruction.

Enabling Technologies (Systems)

- **Secure channels** for distributing shares (TLS/PKI).
- **Reliable randomness** for sampling coefficients/nonces.
- **Distributed storage/replication** across nodes/regions (threat model: partial compromise).
- **Secure hardware** (HSM/TEE) can protect shares-at-rest and limit extraction.

Applications (Where It's Used)

- **Threshold cryptography:** split signing/decryption keys (no single key holder).
- **Secure storage:** distribute shares across servers to reduce breach impact.
- **E-voting:** CRT-based secret sharing constructions in voting/tally workflows [Ift07].
- **Authentication/recovery:** split recovery secrets in MFA-like designs [OBM⁺18].

How to Choose a Scheme

- If policy is "any t of n ": use a threshold scheme (baseline).
- If policy is complex: use a construction for general access structures (watch share size).
- If cheating is realistic: add verifiability or cheating-secure reconstruction.
- If the secret is long-lived: consider proactive refresh.

Takeaways

- Secret sharing formalizes **who can reconstruct** via an access structure $\mathcal{A}$.
- Threshold schemes are the baseline for t -out-of- n policies.
- General policies are possible, but share size and efficiency are the main challenges.
- Real deployments may need extensions: robustness (cheating), proactive refresh, etc.

Thank you!

Questions?

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